

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	<u>Kanmani L, G.Hima Bindu</u>
Register Number	192511414,192511377

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%



Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50



Code of Conduct

I Kanmani L(192511414), G.Hima Bindu (192511377) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Technical Presentation Topics Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. Cloud computing characteristics and benefits for startup ecosystems.
2. How virtualization enabled the growth of cloud service providers.
3. A comparative presentation on IaaS, PaaS, and SaaS for an educational institution.
4. Web services in cloud platforms: SOAP vs REST for enterprise integration.
5. Why XaaS is becoming a dominant consumption model in modern IT.

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

A Comparative Presentation On Iaas, Paas, And Saas For An Educational Institution.

CSA1522-CLOUDCOMPUTING AND BIGDATA ANALYTICS

Guided by:
Dr.P.RAJARAM



Presented by:
KANMANI L(192511414)
G.HIMA BINDU(192511377)



CLOUD COMPUTING IN EDUCATION: OVERVIEW & OBJECTIVES

- **Cloud Computing in Education**

Provides **IT resources and digital services over the Internet.**

Supports **online learning, data storage, student management & collaboration.**

Reduces the need for extensive **physical IT infrastructure.**

- **Objectives**

Understand **IaaS, PaaS & SaaS.**

Compare their **control, management & applications.**

Explore their role in **educational institutions.**

Identify their **practical and industry relevance.**

- **Key Benefits**

Cost Efficiency | Scalability | Anywhere Access | Easy Collaboration

UNDERSTANDING IAAS, PAAS & SAAS

❑ IaaS – Infrastructure as a Service

Provides **virtual servers, storage, and networking**.

Educational Use: Hosts student databases and institutional systems.

Example: Amazon EC2

❑ PaaS – Platform as a Service

Provides a **platform to develop and deploy applications**.

Educational Use: Builds attendance and learning portals.

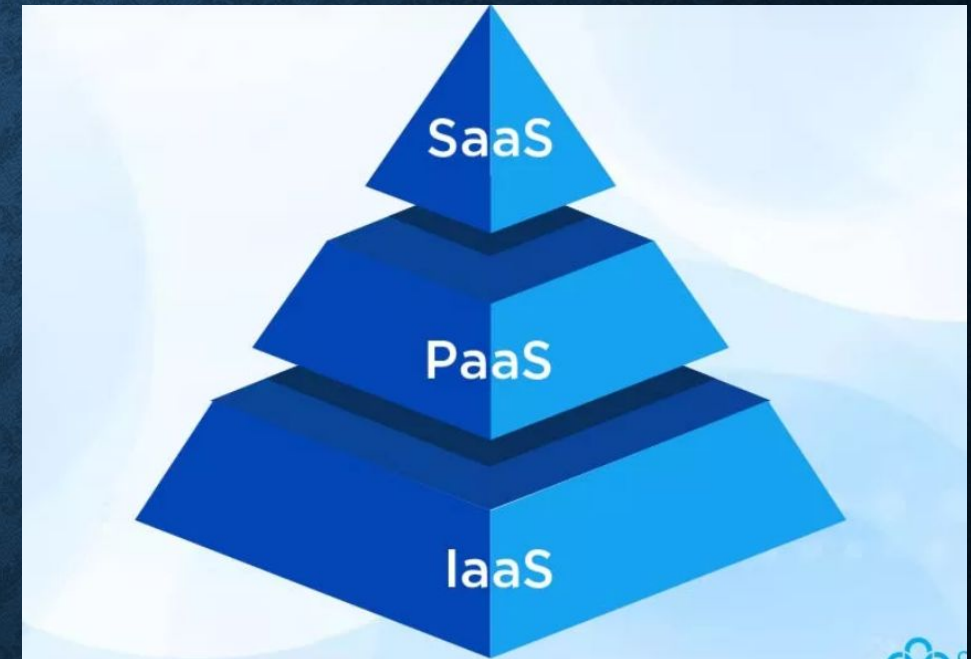
Example: Google App Engine

❑ SaaS – Software as a Service

Provides **ready-to-use software over the Internet**.

Educational Use: Email, collaboration, and learning tools.

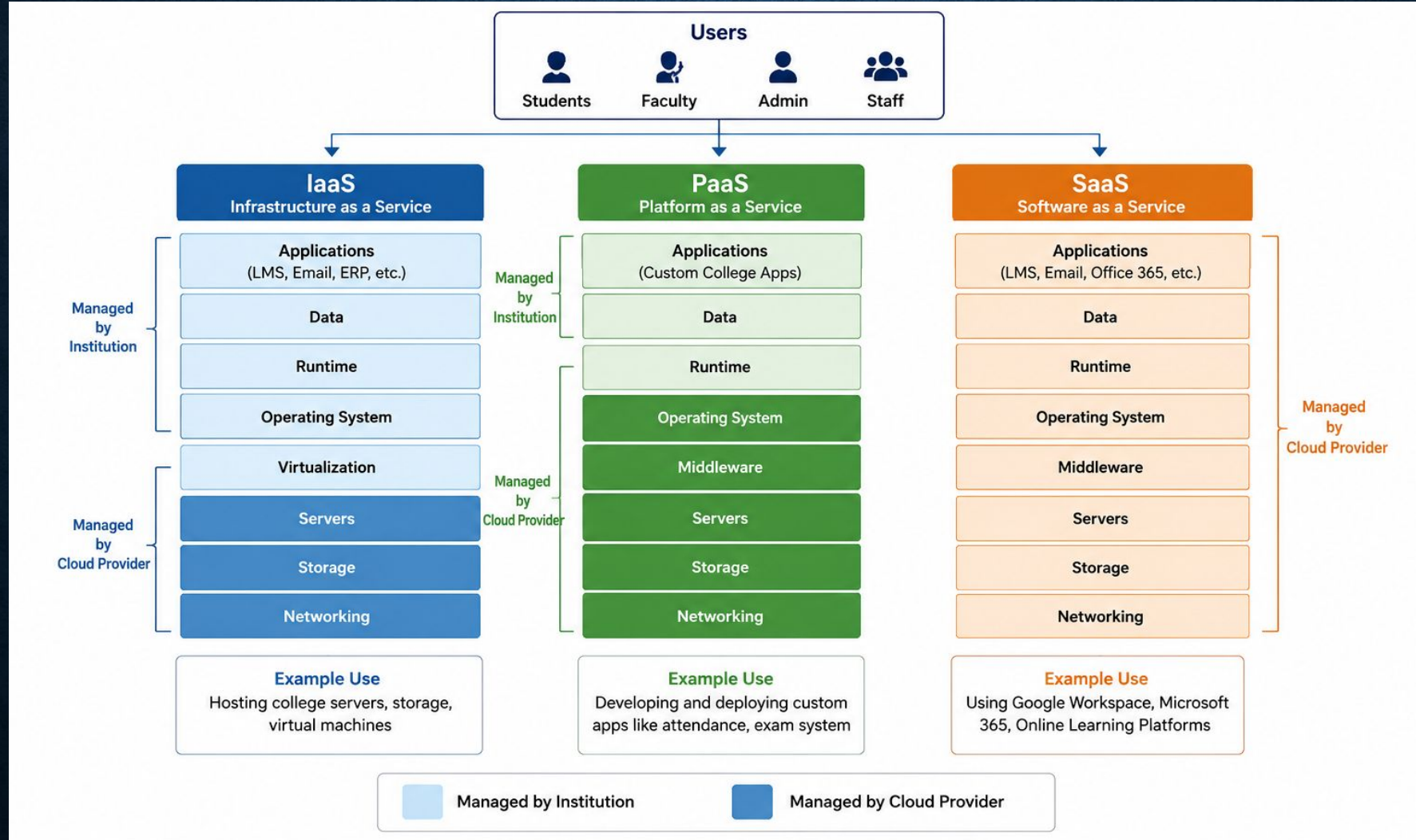
Example: Google Workspace



Core Difference

- IaaS → Infrastructure | PaaS → Application Development | SaaS → Ready-to-use Software

CLOUD SERVICE MODEL ARCHITECTURE



IAAS, PAAS & SAAS IN AN EDUCATIONAL INSTITUTION

❑ IaaS – Infrastructure

- Provides the **IT foundation** for college systems.
Used for: Student databases • Virtual servers • Cloud storage

❑ PaaS – Application Development

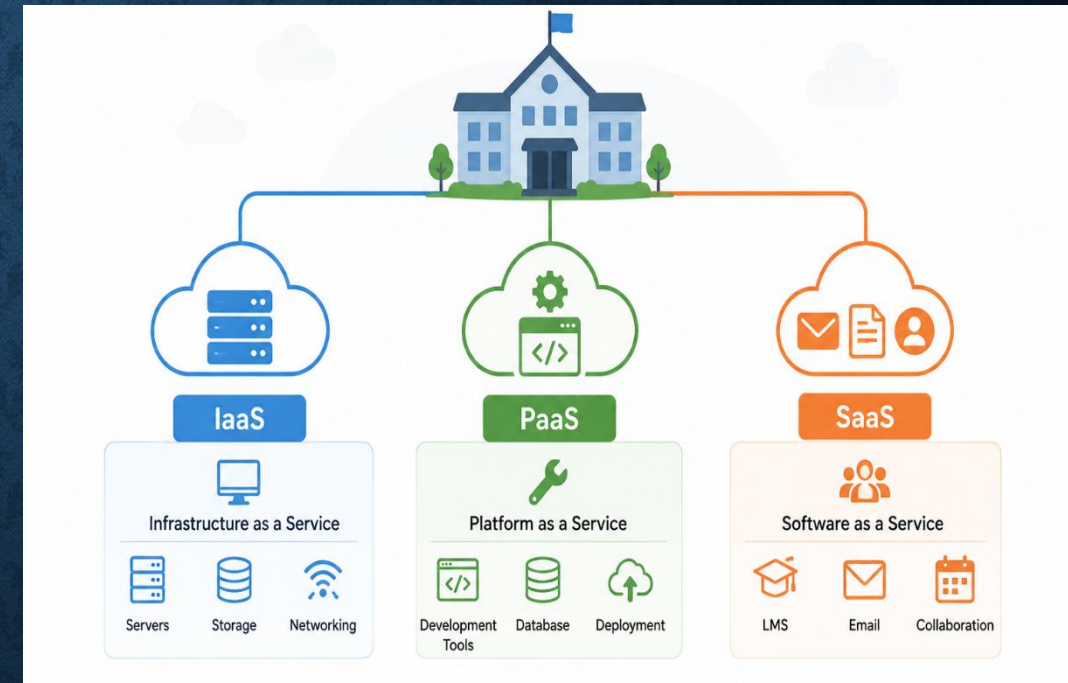
- Provides a platform to **build and deploy custom applications**.
Used for: Attendance systems • Learning Management Systems
- • Student portals

❑ SaaS – Ready-to-Use Applications

- Provides software directly to **students, faculty, and staff**.
Used for: Google Workspace • Microsoft 365 • Online learning tools

❑ Primary Users

- IT Administrators → IaaS | Developers → PaaS | Students & Faculty → SaaS



COMPARATIVE ANALYSIS OF IAAS, PAAS & SAAS

Criteria	IaaS	PaaS	SaaS
Provides	IT Infrastructure	Development Platform	Ready-to-use Software
Management	More user responsibility	Shared responsibility	Mostly provider-managed
Primary Users	IT Administrators	Developers	Students & Faculty
College Application	Host IT systems	Build custom applications	Use digital services
Example	AWS EC2	Google App Engine	Google Workspace

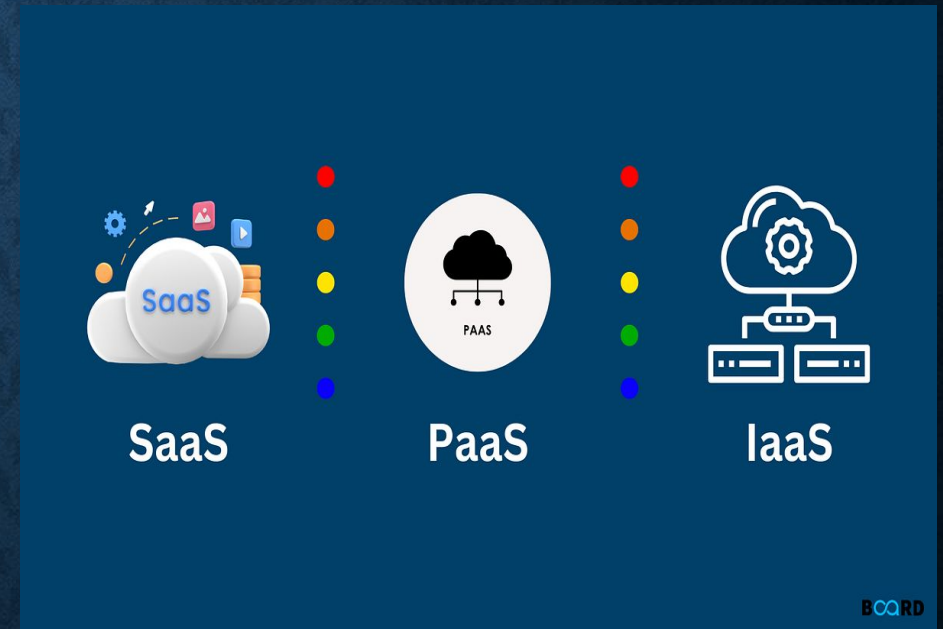
- **IaaS → Maximum Control**
PaaS → Balanced Control & Convenience
SaaS → Maximum Convenience
- **As we move from IaaS → PaaS → SaaS, user control decreases while ease of use increases.**

REAL-WORLD USE CASE: CLOUD-BASED SMART COLLEGE

College Using All Three Cloud Models

- **IaaS – Infrastructure**
Student Information System
→ Stores student records and databases on cloud servers.
- **PaaS – Platform**
Attendance & Learning Portal
→ Helps the college develop and run custom applications.
- **SaaS – Software**
Google Workspace / Microsoft 365
→ Provides email, documents, file sharing, and collaboration.
- **Benefits**

Lower Cost | Scalability | Easy Access | Better Digital Services





INDUSTRIAL RELEVANCE AND CONCLUSION

Industry Relevance

- Supports digital transformation in **education and other industries**.
- Reduces dependence on costly physical IT infrastructure.
- Helps organizations choose services based on **cost, control, and technical needs**.

Conclusion

- **IaaS** is suitable when greater control is required.
- **PaaS** is suitable for institutions developing customized digital solutions.
- **SaaS** is suitable for everyday academic and administrative services.
- A college can **use a combination of all three** to meet different requirements.

Key: The right cloud model depends on an institution's needs, resources, and desired level of control.



THANKYOU

